

$$a_0 = 2$$

$$a_1 = 5$$

$$a_{n+2} = 5a_{n+1} - 6a_n$$

$$a_{n+2} - 5a_{n+1} + 6a_n = 0$$

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \dots$$

$$xf(x) = a_0x + a_1x^2 + a_2x^3 + a_3x^4 + a_4x^5 + \dots$$

$$x^2f(x) = a_0x^2 + a_1x^3 + a_2x^4 + a_3x^5 + a_4x^6 + \dots$$

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \dots$$

$$xf(x) = a_0x + a_1x^2 + a_2x^3 + a_3x^4 + \dots$$

$$x^2f(x) = a_0x^2 + a_1x^3 + a_2x^4 + \dots$$

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \dots$$

$$-5xf(x) = -5a_0x - 5a_1x^2 - 5a_2x^3 - 5a_3x^4 - \dots$$

$$6x^2f(x) = 6a_0x^2 + 6a_1x^3 + 6a_2x^4 + \dots$$

$$\begin{aligned} f(x) - 5xf(x) + 6x^2f(x) &= a_0 + a_1x - 5a_0x = [0 \text{ osadit}] \\ &= 2 + 5x - 10x = \\ &= 2 - 5x \end{aligned}$$

$$f(x) = \frac{2 - 5x}{1 - 5x + 6x^2}$$

$$f(x) = \frac{2 - 5x}{(1 - 2x)(1 - 3x)} = \frac{A}{1 - 2x} + \frac{B}{1 - 3x}$$

$$2 - 5x = A(1 - 3x) + B(1 - 2x)$$

$$\rightarrow A = 1, B = 1$$

$$f(x) = \frac{1}{1 - 2x} + \frac{1}{1 - 3x} = \sum 2^n x^n + \sum 3^n x^n$$

$$\boxed{a_n = 2^n + 3^n}$$